

Design FMEA (qualitative version) — VBF-T1R1-DFMEA-01

Case: t1_r1 | Date: 2026-08-25 | Prepared: _____ | Reviewed: _____ | Approved: _____

Qualitative version, based on simulation signals (S/O: high/medium/low; basis = magnitude of the simulated value vs threshold).
RPN = simplified S×O qualitative matrix (annotated qualitative caliber).

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Negative electrode plating (fast charge)	Anode surface potential < 0 V vs Li/Li+ during 4C CC charge from deep discharge (transport/kinetic polarization at 20 A)	cell/r3_l_4c_dfn.json:anode_potential_v min = +0.0249 V (24.9 mV margin; round-1 baseline was -0.4385 V)	high	low	Implemented: graphite particles 2.5 μm, anode porosity 0.40 (120 μm), separator 6 μm, sigma = 1.8 S/m, t+ = 0.55, liquid cooling h = 100. Residual note: formula-caliber N/P = 0.752 (usable-window ≈ 1.0) — no N/P headroom; plating safety relies on kinetics/transport margin. Recommend: pack-level 4C charge with CV phase before production.
Thermal runaway risk (temperature exceedance)	Cell temperature above 60 °C red line under 4C charge	cell/r3_l_4c_dfn.json:T_max_K = 325.655 K vs 333.15 K (margin 7.50 K); overcharge TR triggered = false (cell/r3_l_tr.json)	high	low	Implemented: liquid cooling h = 100 W/m2K; high-conductivity electrolyte cuts ohmic heat. Recommend: verify margin under pack-level insulation; monitor CV-phase charging heat.
Electrolyte oxidative decomposition (voltage window)	Electrolyte HOMO/oxidation limit vs cathode potential	Not assessed this case: molecular-layer criteria (max_homo_ev) unchecked, real_compute = false (no DFT)	high	medium (unverified — honest)	Not yet implemented. Recommend: Stage-5 true DFT endorsement (HOMO level vs 4.2 V-class cathode) before production; overcharge to 4.7 V already simulated without TR (cell/r3_l_oc.json) but does not prove long-term electrolyte stability.
Insufficient capacity	Design capacity below target	cell/r3_l_1c.json:capacity_ah = 5.6505 Ah vs parameter nominal 5.0 Ah (no registered capacity criterion)	medium	low	Implemented: capacity verified at 1C. Round-2 lesson: anode AMVF 0.65 became near-limiting (capacity -2.4%); round-3 thickness compensation (120 μm) restored margin — retained in finalist.

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Anode-limited discharge (capacity loss)	Anode active-material share too low after porosity increase	round-2 evaluate: 1C capacity 5.743 -> 5.602 Ah at AMVF 0.65	medium	low	Implemented: anode thickness 110 -> 120 μm (params_r3_l.json); finalist 1C capacity 5.6505 Ah.

Conclusion

Highest-risk items: electrolyte oxidative stability (unverified at molecular level — recommend Stage-5 DFT endorsement before production) and plating-margin erosion at pack level (recommend CV-phase and pack-thermal verification). All simulation-signal items with registered criteria have mitigation implemented and verified in the finalist (r3-l). Complete FMEA including process/supplier failure modes: N/A (beyond pure simulation boundary).